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$$\int \frac{\sin x dx}{(1+3\cos^2 x)\cos x}$$

$$u = \cos x$$

$$du = -\sin x dx$$

$$\Rightarrow -\int \frac{1}{(1+3u^2)u} du$$

$$= -\int \left(\frac{A}{u} + \frac{Bu+C}{1+3u^2}\right) du$$

$$A = \left.\frac{1}{1+3u^2}\right|_{u=0} = 1$$

$$1+3u^2 = 0$$

$$\Leftrightarrow 3u^2 = -1$$

$$\Leftrightarrow u^2 = -\frac{1}{3}$$

$$\Leftrightarrow u = \frac{i}{\sqrt{3}}$$

$$B\frac{i}{\sqrt{3}} + C = \left.\frac{1}{u}\right|_{u=\frac{i}{\sqrt{3}}} = \frac{1}{\frac{i}{\sqrt{3}}}$$

$$B\frac{i}{\sqrt{3}} + C = \frac{\sqrt{3}}{i}$$

$$B\frac{i}{\sqrt{3}} + C = -\sqrt{3}i$$

$$\Rightarrow (B, C) = (-3, 0)$$

$$\Rightarrow -\int \left(\frac{1}{u} - \frac{3u}{1+3u^2}\right) du$$

$$= -\int \frac{1}{u} du + \int \frac{3u}{1+3u^2} du$$

$$= -\ln|u| + \int \frac{3u}{1+3u^2} du$$

$$v = 1+3u^2$$

$$dv = 6udu$$

$$\frac{dv}{2} = 3udu$$

$$\Rightarrow \int \frac{3u}{1+3u^2} du = \frac{1}{2} \int \frac{1}{v} dv$$
$$= \frac{1}{2} \ln |v|$$
$$= \frac{1}{2} \ln |1+3u^2|$$
$$\Rightarrow \int \frac{\sin x dx}{(1+3\cos^2 x)\cos x} = -\ln |u| + \frac{1}{2} \ln |1+3u^2| + C$$
$$= -\ln |\cos x| + \frac{1}{2} \ln |1+3\cos^2 x| + C$$

References